LAPORAN PRAKTIKUM

SISTEM OPERASI

PERTEMUAN 4

(MEMBACA SOURCE CODE XINU)

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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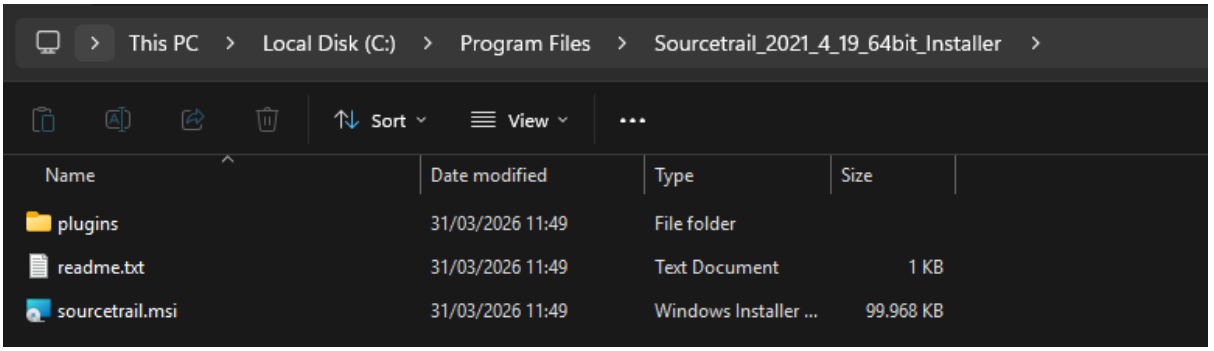
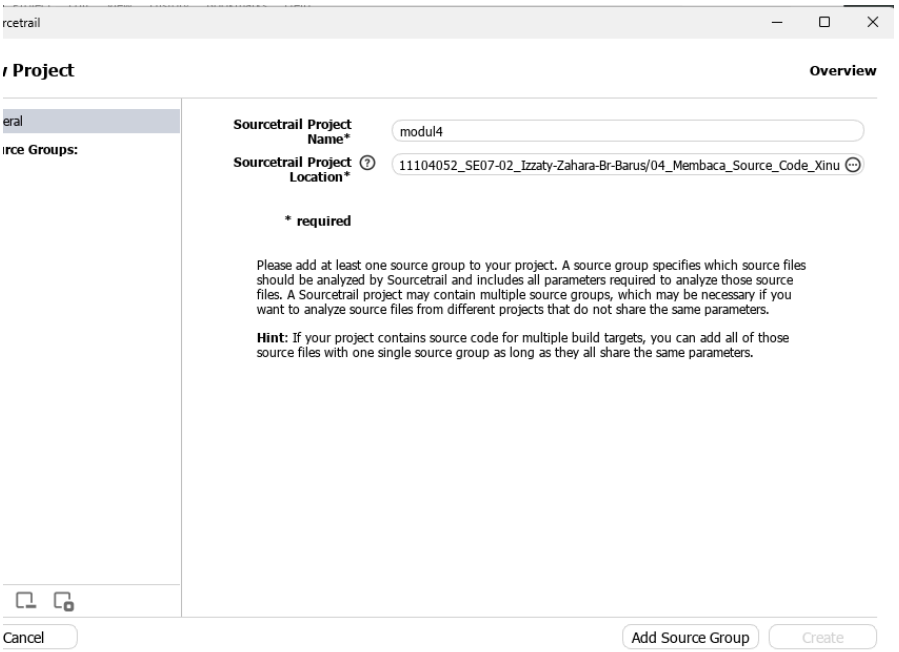
Jurnal

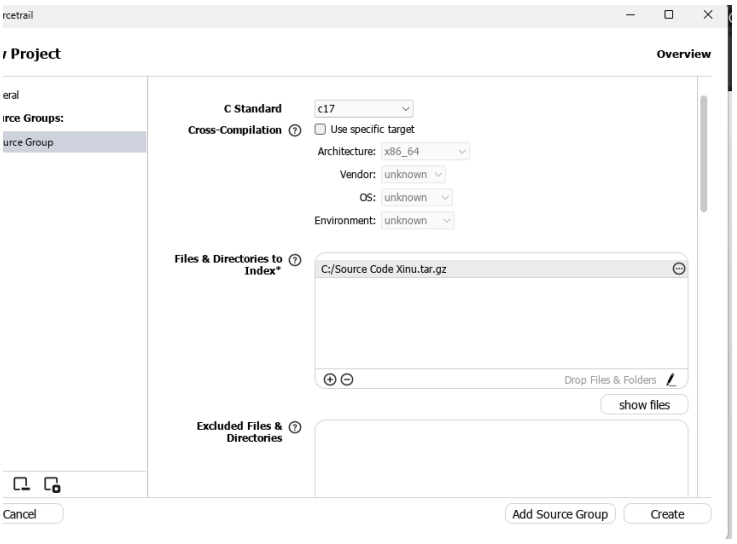
1. Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut? Hint: baca kembali modul-modul sebelumnya
   =**Nama image:** xinu.elf (atau xinu.boot tergantung konfigurasi platform)
   Berdasarkan materi modul, output dari proses cross-development adalah sebuah **file image**. Pada Xinu, setelah menjalankan perintah make di dalam direktori xinu/compile, file image yang dihasilkan bernama:**Lokasi file:** berada di direktori xinu/compile/
   Ukuran file: bervariasi tergantung konfigurasi, namun umumnya berkisar antara **300 KB – 1 MB**.
2. Membaca source code Xinu
   a. Cek aplikasi bernama Sourcetrail di PC. Jika belum ada, download SourceTrail pada (DOWNLOAD SOURCETRAIL). SourceTrail adalah software untuk mengeksplorasi source code. Programmer yang bagus lebih banyak membaca kode daripada menulis kode

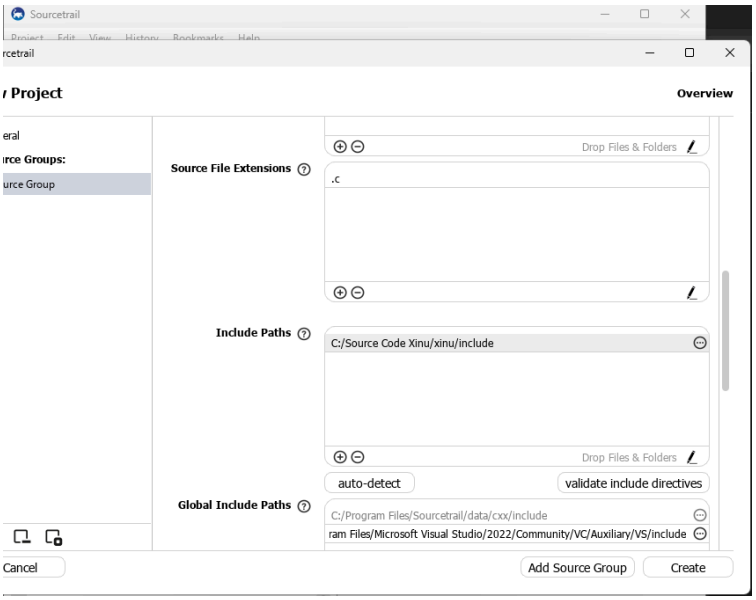


   b. Project → New Project

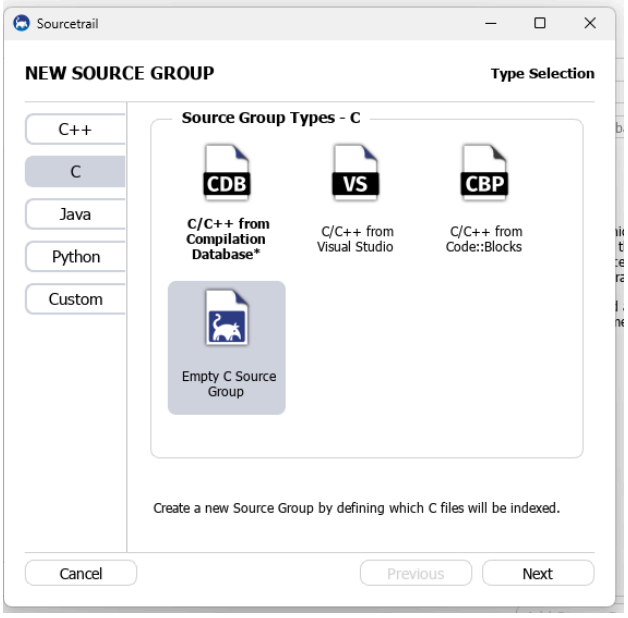


c. Isi nama project xinu dan pilih lokasi project di manapun

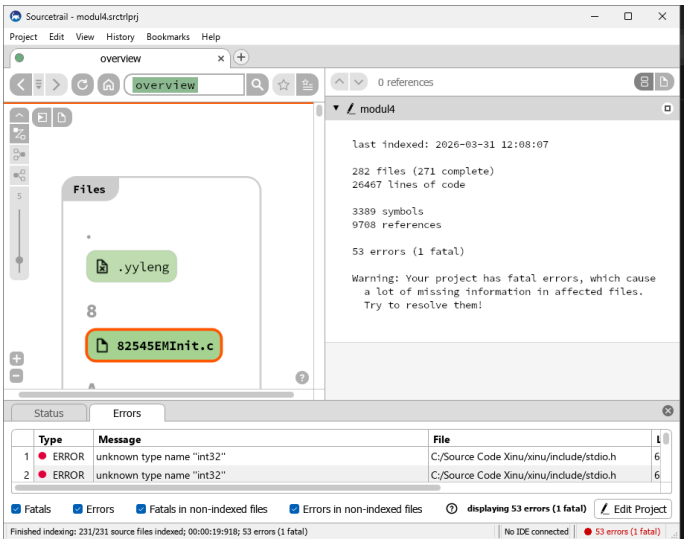


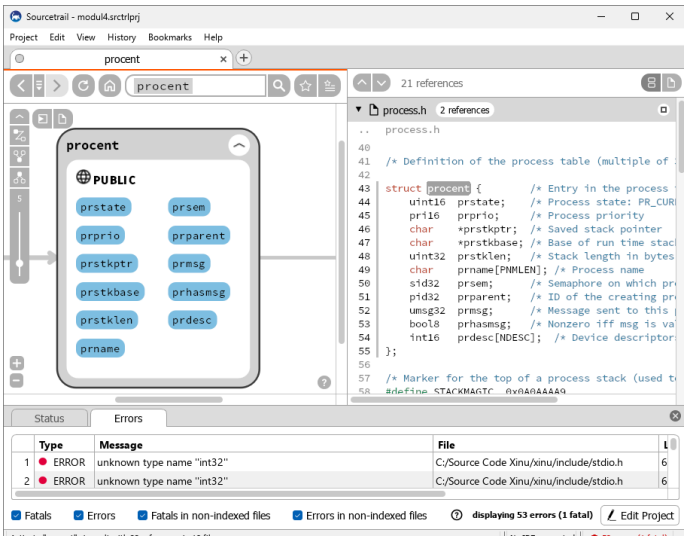


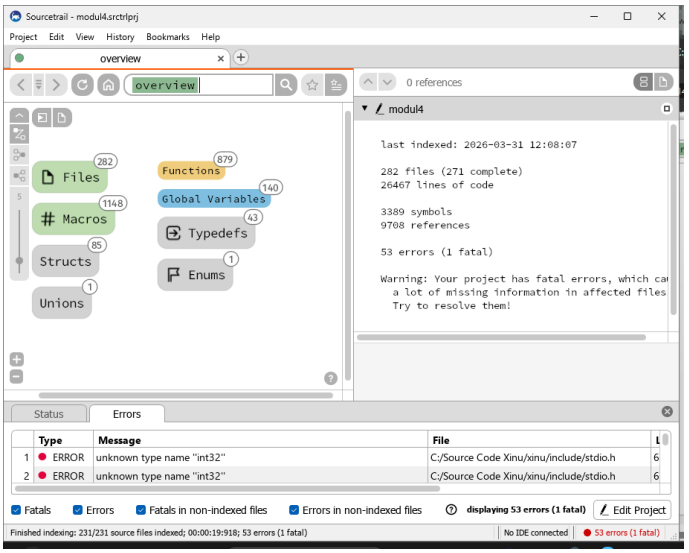
d. Add Source Groups, pilih C, lalu pilih Empty C Source Group

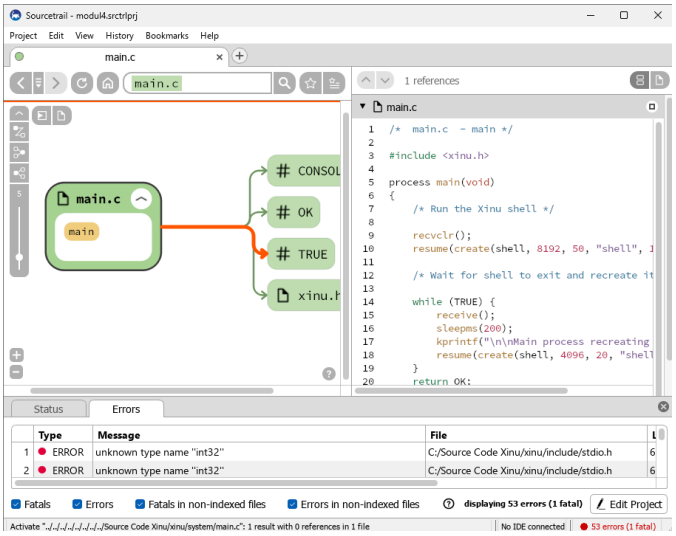


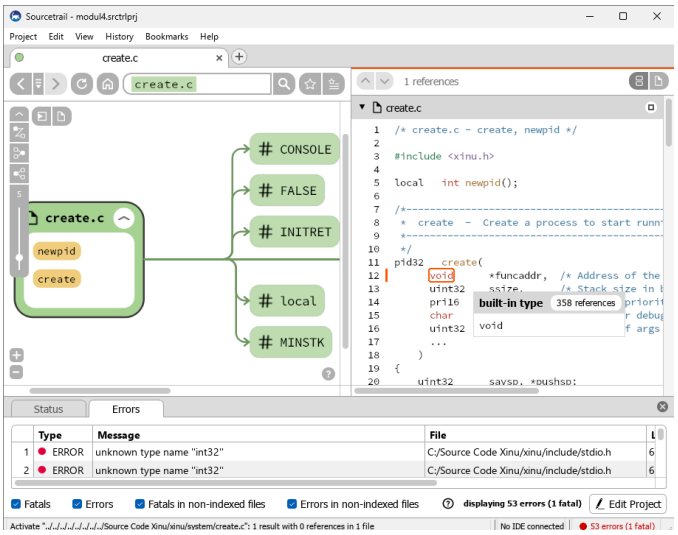
f. Silahkan eksplorasi source code Xinu

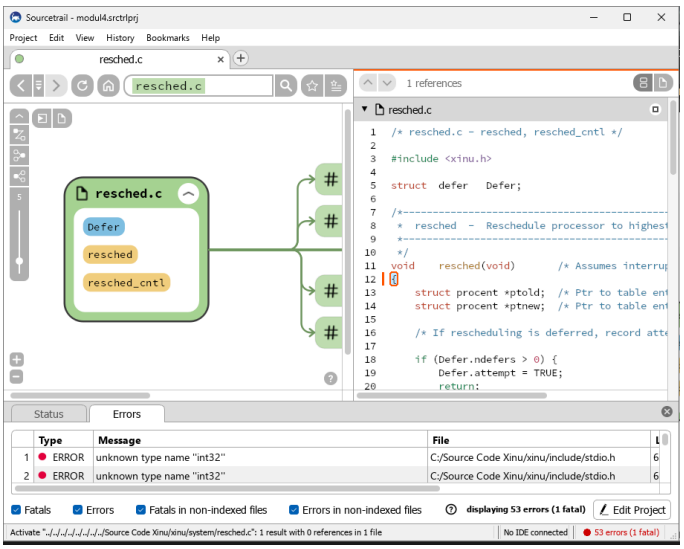












3. Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut? Hint: file berektensi .h Pada Linux bisa digunakan perintah “grep” Contoh: $ grep -r “kata_yang_ingin_dicari” /path/ke/direktori/tempat/file/berada
    a. Struktur data proses.h

```
xinu@xinu-develop-end:~$ ls
prev-version  README  xinu
xinu@xinu-develop-end:~$ pwd
/home/xinu
xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
xinu@xinu-develop-end:~$
```

b. perintah “grep -r "struct procent" /home/xinu/xinu/include/”

```
/home/xinu
xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/include
/home/xinu/xinu/include/shell.h:/* Shell banner (assumes VT100) */
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/shell/
/home/xinu/xinu/shell/shell.c:  /* Print shell banner and startup message */
xinu@xinu-develop-end:~$ grep -r "struct procent" /home/xinu/xinu/include/
/home/xinu/xinu/include/process.h:struct procent {              /* Entry in the process
 table          */
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC           8
#endif

/* Process state constants */
```

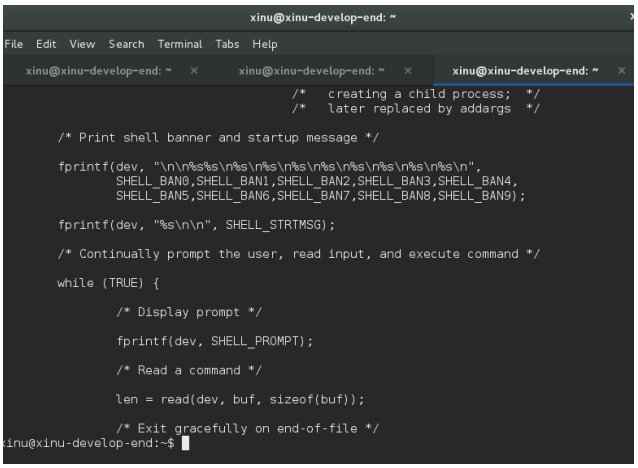
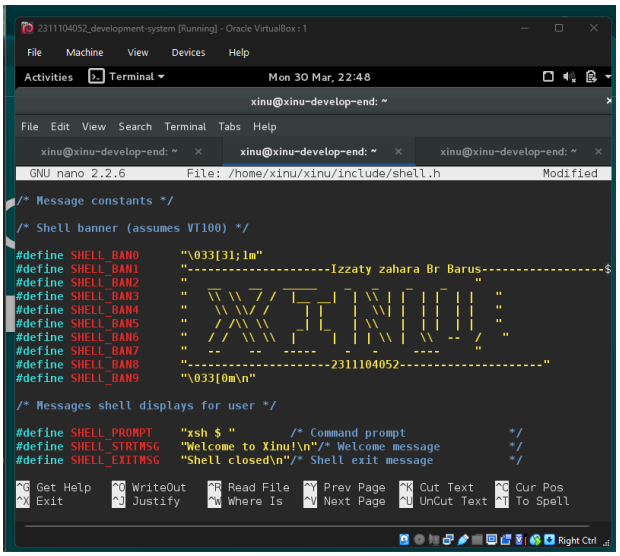
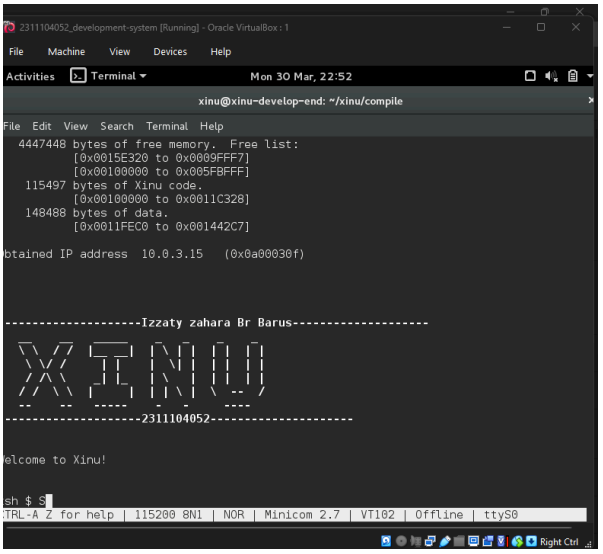
```
* Process state constants */

define PR_FREE          0       /* Process table entry is unused        */
define PR_CURR          1       /* Process is currently running         */
define PR_READY         2       /* Process is on ready queue            */
define PR_RECV          3       /* Process waiting for message          */
define PR_SLEEP         4       /* Process is sleeping                  */
define PR_SUSP          5       /* Process is suspended                 */
define PR_WAIT          6       /* Process is on semaphore queue        */
define PR_RECTIM        7       /* Process is receiving with timeout    */

* Miscellaneous process definitions */

define PNMLEN           16      /* Length of process "name"             */
define NULLPROC         0       /* ID of the null process               */

* Process initialization constants */

define INITSTK          65536   /* Initial process stack size           */
define INITPRIO         20      /* Initial process priority             */
define INITRET          userret /* Address to which process returns     */

* Inline code to check process ID (assumes interrupts are disabled)    */

define isbadpid(x)      ( ((pid32)(x) < 0) || \
```

4. Mengubah welcome banner pada Xinu

a. Carilah file yang menyimpan banner Xinu! Hint: file berekstensi .h pada direktori xinu/include

```
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/include/
/home/xinu/xinu/include/shell.h:/* Shell banner (assumes VT100) */
xinu@xinu-develop-end:~$
```

b. Carilah file yang menampilkan banner Xinu! Hint: file berektensi .c pada direktori xinu/shell



c. Ubahlah welcome banner Xinu